

Metadata

Alpha name

Author

Date

Universe

Neutralization

Decay

Truncation

Alpha Design Steps

Step	Guiding prompts	Your answer
1	Define objective & edge What market inefficiency do you believe exists? Why has it not been fully arbitrated? What outcome do you expect (e.g. mean reversion, drift)?	
2	Formulate hypothesis If X changes, what should happen to returns? Clearly define Long and Short positions. What is the expected holding horizon?	
3	Mathematical expression List all raw variables used. Define units, scaling, and data treatment. Write the base formula before transformations.	
4	Transform & refine Choose operators: z-score, rank, group-rank, neutralization. Specify lookback windows, decay, truncation. Expected direction: high = long or short?	
5	Robustness & final alpha Key risks or failure cases. Suggested robustness checks (subperiods, sectors, turnover). Final alpha formula (pseudo-code) and rationale.	

1. Objective & Edge

Short-term price dislocations arise from liquidity shocks and investor overreaction. These effects are temporary and tend to mean-revert.

2. Hypothesis

Stocks with extreme negative short-term returns will outperform, while extreme positive return stocks will underperform.
Expected holding horizon: 2–10 trading days.

3. Mathematical Expression

Raw variable: daily returns.
Compute cumulative returns over a short window (e.g. 5 days).

4. Transform & Refine

Apply cross-sectional z-score to identify extremes.
Rank within industry to control sector exposure.
Low z-score = Long, High z-score = Short.

5. Final Alpha Formula

Pseudo-code:
`group_rank( -zscore(ts_sum(returns, 5)), industry )`

Rationale: exploits short-term mean reversion while neutralizing industry effects.

Alpha Design – Multi-Alpha Work Table

Each row = one alpha idea

Alpha						